

YIXUAN LI

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Education

Southern University of Science and Technology

Sep. 2022 – Present

Undergraduate in Data Science

Shenzhen, China

- **GPA:** 3.79/4.00 (Junior Year: 3.92/4.00)
- **Mathematics Courses:** Linear Algebra, Mathematical Analysis, Discrete Mathematics and Its Applications, Foundation of Probability Theory, Mathematical Statistics, Statistical Linear Models, Operational Research and Optimization
- **CS Courses:** Introduction to Computer Programming, Data Structures and Algorithm Analysis, Principles of Database Systems, Distributed Storage and Parallel Computing, Advanced Natural Language Processing, Artificial Intelligence

University of California, San Diego

Mar. 2025 – Jun. 2025

University and Professional Studies (UPS) Program

San Diego, CA, United States

- **GPA:** 4.00/4.00
- **Courses:** Machine Learning Algorithms, AI Search and Reasoning, Topics of Data Science: Trustworthy Machine Learning

Research Interest

- Trustworthy Machine Learning, Uncertainty Quantification, Interpretable AI, Hallucination Detection
- AI for Science (Biology), AI for Health, Medical Imaging

Research & Professional Experience

Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction

Oct. 2025 – Present

Undergraduate Research Assistant — Advisor: Prof. Hongxin Wei, Prof. Shimin Shuai

SUSTech, China

- Developed a UNet-based model to reconstruct spatial maps from RNA transcriptomic data and predict protein expression levels at spatial coordinates.
- Achieved an Average Spearman Correlation of **0.65** on the test set, ranking **Top 1** on the Leaderboard.
- Enabled accurate spatial proteomic prediction and provides a foundation for downstream biological analyses.
- A private GitHub repository will host the implementation (due to participation in an official challenge).

Statistical Analysis of Mental Health Impact among Menopausal Women

Sep. 2025 – Present

Undergraduate Research Assistant — Advisor: Prof. Jianqing Shi

SUSTech, China

- Analyzed large-scale survey data on menopausal women in Malaysia to examine the relationship between menopausal stages and mental health outcomes.
- Applied statistical methods including Kruskal-Wallis, Fisher-Exact, Chi-square, ANCOVA, CMH, Paired t-tests, and multiple imputation for hypothesis testing and subgroup analysis.
- Identified stage-adjusted psychological risk factors and enhanced inference robustness in women's health research.

Benchmarking Uncertainty Quantification Methods in Large Language Models

Feb. 2025 – Present

Undergraduate Research Assistant — Advisor: Prof. Hongxin Wei

SUSTech, China

- Built a unified benchmarking pipeline for 10+ semantic uncertainty quantification methods (e.g., SE, DSE, SAR) on QA datasets using open-source LLMs, evaluated via AUROC, AUARC, and PRR.
- Standardized data processing and evaluation for reproducible comparison across uncertainty estimation approaches.
- Conducted an extensive literature review on semantic uncertainty in LLMs, connecting with hallucination detection and calibration, and published findings in technical blog posts.

Machine Learning for Clinical Decision Support in Orthodontics

Jun. 2024 – Aug. 2024

Summer Internship — Mentor: Dr. Ying Shan

Clinical Research Institute, PKUSZ Hospital

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which involved cleaning, de-duplication, and standardization.
- Developed linear and neural network models based on 16 craniofacial measurement variables to predict patient-specific tooth extraction plans.
- Achieved promising prediction accuracy and facilitated internal adoption to support orthodontic treatment planning.

Projects & Competitions

Concept Bottleneck in Generative Models: Reproduction Study | Core Member

Apr. 2025 – Jun. 2025

- Reproduced the CVPR paper on Post-hoc Concept Bottleneck frameworks: *Concept Bottleneck Autoencoder (CB-AE)*, *Concept Controller (CC)* using *StyleGAN* framework, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on *Concept Extra-Interpolation*, *Entanglement*, and *Leakage* to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including Github code repository, presentation slides, and a public technical blog, achieving a course score of **43.4/40 (Top 1)** under the supervision of Prof. Lily Weng.

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | Team Leader

Oct. 2024 – Jan. 2025

- Led a team to fine-tune *Gemma-2B* on the *GPT-4 Chinese QA dataset*. Built a complete fine-tuning pipeline covering data preprocessing, LoRA integration, and evaluation. Published notebook can be viewed [here](#).
- Integrated *Retrieval-Augmented Generation (RAG)* for data augmentation and *LoRA-based PEFT* for parameter-efficient fine-tuning, resulting in improved dataset diversity, contextual grounding, and instruction-following performance.
- Evaluated model performance using *BLEU*, *ROUGE*, *METEOR*, and *BERTScore metrics*, and performed *cross-lingual robustness analysis* showing stable results across four languages.

Reinforcement Learning on FrozenLake (Course Project) | Core Member

Oct. 2024 – Jan. 2025

- Implemented algorithms (*Double DQN*, *Dueling DQN*), trained and analyzed agents on FrozenLake environment.
- Performed *hyperparameter tuning* and *robustness comparison* across algorithms, and achieve a performance with **98%** accuracy.
- Compiled a comprehensive project report and structured a well-documented GitHub repository to ensure reproducibility.

Awards

- **First-class Scholarship for Outstanding students, SUSTech** (Top 5%)

2023 – 2024, 2024 – 2025
- **Third-class Scholarship for Outstanding students, SUSTech** (Top 15%)

2022 – 2023
- **Second Prize** Awarded in the 2024 China Undergraduate Mathematical Contest in Modeling (CUMCM), in Guangdong division

Oct. 2024
- **Honorable Mention** Awarded in the 2024 Mathematical Contest In Modeling (MCM)

May 2024
- **Excellent Student Award** For outstanding students in SUSTech for three times

2022 – 2025

Leadership / Extracurricular

Community Involvement

Oct. 2023 – Present

Volunteer

SUSTech

- Participated in online teaching programs, delivering educational lessons to children in rural areas.
- Accumulated over **179 hours** of volunteer service over three years.
- Recognized as an **Outstanding Volunteer** at both the university and School of Science levels.

Media of Shuren College, SUSTech

Sep. 2023 – Aug. 2024

President

SUSTech

- Leading a team with 15 Team members. Created numerous contents/posts for the college’s official media account.
- Received the **Excellent College Student Leader Award** and the **Outstanding College Organizations Award**.

Pickleball Team

Oct. 2023 – Present

Member

SUSTech

- Represented SUSTech in the Second Guangdong Pickleball Competition (Women’s Doubles, **Top 16** finish)
- Participated in the 2023 SUSTech Pickleball Championship, achieving **5th place** in Mixed Doubles.

Skills

- **Programming:** Python, R, SQL, Java, JavaScript, HTML, Hadoop
- **Development Tools:** PyTorch, Git, Jupyter Notebook, LaTeX, Markdown
- **Languages:** English (Fluent, TOEFL 103, GRE 322), Mandarin (Native), Cantonese (Fluent)